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Module : Diagram Explanation

Start with USER : Left side of your diagram :

[Laptop / Mobile]

The person using the mobile to access the application , For example you open

https://electricitybill.com

And can use : Chrome , Firefox , Mobile Browser , Postman and another application

The browser or application is called the client

Client is the thing asking the server for something for example : You open :

https://electricitybill.com/login

Your browser sends a request like “Give me the login page “ and browser is the client and another example like our react application sends :

GET /api/bills

The react application acts as a client

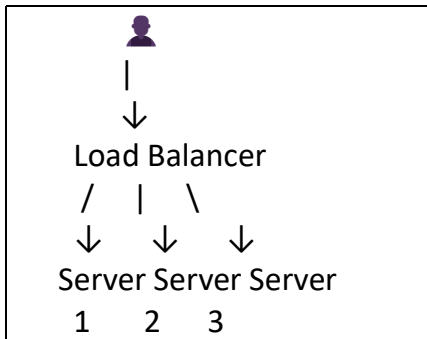
HTTPS / HTTP is the communication between the client and server and client wants his electricity bills through the server and also this the language / rules used to send that request

http://example.com

https://example.com

HTTPS is the secure encrypted version and in real applications we normally use HTTPS

AWS Concept : Load Balancer : [Diagram has APP as ALB] :



Imagine a user uses load balancer for using 3 different servers and the load balancer receives the users request and decides which server should handle this request

Suppose you have only one server and the server may become overloaded and so now traffic can be distributed and this is called load balancing

User 1	→	Server 1
User 2	→	Server 2
User 3	→	Server 3
User 4	→	Server 1
User 5	→	Server 3

ALB is called Application Load Balancer and in AWS ALB works using EC2 servers and it works at application and HTTP level and can understand things such as

/api/users
/api/bills
/login
/products

Application servers run spring boot application using different EC2s

Suppose your spring boot application has

GET /api/bills
POST /api/login
GET /api/users

The application server runs the code that handles these requests when a request arrives such as

GET /api/bills

And spring boot process it

Inside a Server : Right side of diagram :

These are layers :

Java/Python/JavaScript/Go/Ruby
Libraries/Frameworks/SDK
Compile or Interpret
Runtime
OS
Network + CPU + RAM

CPU + RAM + Network

CPU	CPU is the computer processing brain and CPU executes instructions
RAM	RAM is temporary working memory as application needs memory so its uses RAM
Network	Network Allows the server to communicate with users , servers , database and AWS services

Operating System

Operating System consists of Ubuntu Linux , Amazon Linux , and Windows Server as AWS environments work with Linux

Runtime

Suppose your application is written in java and server needs something capable of running java and this called java runtime

For java application we use JVM and Python application we use Python runtime and for JAVA Script and Node.js we use Node.js runtime

Frameworks : Diagram : [Libraries / Frameworks / SDK] :

A framework gives you ready made tools for building applications such as for Java we use spring boot and Java script we use react or node.js and for python we use Django / Flask / Fast API

Instead of building everything from scratch you use frameworks

Library

A library uses reusable code instead of writing everything you use libraries for application with spring libraries and Java

SDK

Known as Software Development Kit where it provides tools to communicate with a platform or service

Your application uses AWS SDK to communicate with Amazon S3 so Spring boot uses AWS SDK to use S3

Compile : Diagram : Compile or interpret :

This is about converting your application code into something the computer can execute

Data Storage : Bottom right corner of your diagram :

Where application data lives and Electricity Bill application may store :

Customer : Bill : Meter : Payment : Username : Password

Database

PostgreSQL and MYSQL are database systems and spring boot uses MYSQL and based for customer we get :

id : name : email : address

Another table for bills we get :

id : customer_id : amount : date : status

AWS RDS

In AWS instead of installing MYSQL manually on an EC2 server , we can use Amazon RDS

So spring boot uses EC2 with amazon RDS to connect MYSQL and database is managed separately from your application server

Object Storage : Diagram shows : AWS S3 / BLOB Storage

This is different from a database and database is good for structured information such as customer , bill , payment and account

And S3 is good for files such as Images , PDF , Videos , Backups and Documents

We can store PDF document inside Amazon S3

Network Load Balancer : NLB : Diagram shows : Load balancer (Network – NLB)

AWS has different types of load balancers and we mostly use ALB known as Application Load Balancer which works mainly with HTTP / HTTPS application traffic

Network Load Balancer works at the network and transport level and designed for TCP / UDP / TLS use case

So you don't need both ALB and NLB for every application

S3 + CloudFront for React :

React produces frontend files such as index.html , java script files , CSS images and Image and don't need an EC2 server and you can put the built react application into Amazon S3

Then CloudFront distributes it globally

